



news

[08. mar 2023]: SpecEmu was updated.
[12. jan 2021]: SpecEmu was updated.
[16. nov 2019]: SpecEmu was updated.
[08. may 2018]: Links were corrected.
[31. jan 2018]: TBIOS v1.4 sources, DithvIDE and Divideo examples for TBIOS were added.
[23. jan 2018]: SpecEmu emulator was added.
[17. jan 2018]: FATware 0.14a was added.
[16. jan 2018]: New divIDE home started :-)

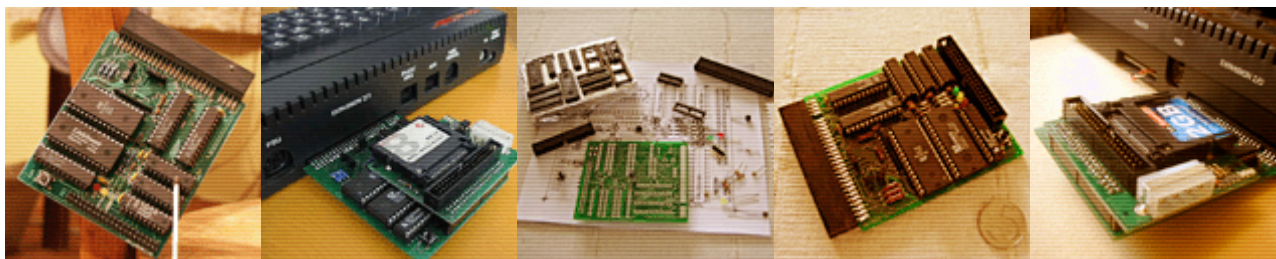
introduction

divIDE is an ATA (IDE) interface by Pavel Cimbal, which takes your ZX Spectrum computing to a whole new level. As the time takes its toll on ageing media such as audio tapes or diskettes, many ZX Spectrum enthusiasts simply stick to emulation for better reliability and comfort. With divIDE you can put your software collection to a hard disk, CD-ROM or even CompactFlash card and experience your favourite games, demos and utilities the way they were meant to be run - and even better. Using existing firmwares, many users find their Speccy box again a live platform - time for you to make the switch? ;]

features

- divIDE uses full 16 bits of ATA bus [\(1\)](#)
- divIDE works on all ZX Spectrum flavours (16, 48, 48+, 128, +2, +2A, +3 and clones)
- Thanks to divIDE's onboard logic, theoretical transfer speed is 218 KB/sec (determined by the latency of INI/OUTI instructions) [\(2\)](#)
- divIDE has 8 KB of shadow flash ROM that hosts operating system core, leaving your original ZX Spectrum ROM intact. Additional 32 KB of RAM accessible as 8 KB memory banks are present.
- divIDE's auto-mapping feature transparently maps shadow ROM at important entry points, enabling standard tape emulation, BASIC extensions, NMI menu and DISCiPLE/+D or BetaDisk emulation.
- divIDE's MAPRAM feature helps developers and users to test new software without the need of reflashing their working system in shadow ROM. It can also emulate another 8 KB of ROM if necessary.
- divIDE works with all ATA-compatible devices (there are no known compatibility issues) [\(3\)](#)
- Available software supports widely used emulator formats (TAP, SNA, Z80, SCR). Just download your all-time favourite games from the web and play!

screenshots



software

Firmwares:

- **FATware 0.14** natively supports up to 8 standard FAT-16 partitions (including long file names). It loads TAP, SNA, Z80, SCR and interlaced SCR files. With FATware you don't need any low-level disk utilities to get up and running - just unplug your disk or CompactFlash from your PC and plug it to your divIDE. NOTE: FATware is read only.
- **ESXDOS** aims to be the ultimate firmware for the DivIDE/DivMMC interface. It supports HDD/CDROM/ZIP/LS120 devices, has full FAT16/FAT32 read/write support (no extended partitions, no LFN), Betadisk/TR-DOS emulation using (trimmed) .TRD files and provides a lot of handy features.
- **DEMFIR (DTP's Emulator Files Runner)** handles ISO 9660 file system found on CD-ROMs. You can use either CDs or unfragmented ISO images stored on a disk. DEMFIR supports TAP, SNA, Z80, MFC and SCR files. You can download these from numerous software archives such as [World of Spectrum](#). As it is possible to fit thousands of games to a single CD, DEMFIR can turn your Speccy into huge retro-gaming console. NOTE: DEMFIR is read only.
- **MDOS3** is a divIDE enhancement of MDOS/MDOS2 systems which were used in D40/D80 disk units. MDOS3 is a full-featured OS working with raw diskette images and supporting up to 4 virtual drives. One of its divIDE-specific add-ons is a tape emulator. With huge D40/D80 software base available this is worth a try, especially if you're a former Didaktik user.
- **+DivIDE** is an adaptation of the GDOS/G+DOS system found in the DISCiPLE/+D disk interfaces. It should work with all DISCiPLE and +D software which doesn't call routines in GDOS/G+DOS ROM directly. +DivIDE uses virtual disks of 1600 sectors each on the ATA drive. Certain GDOS/G+DOS features are extended and improved. NOTE: +DivIDE uses raw LBA access to disk images so take care if you're hosting another file system on your disk.
- **Goldfinch** is an open source project for advaced users. Its goal is to build software for accessing mass-storage on ZX Spectrum using modular dynamic linking mechanism. It's in early stage of development, but looks promising.
- **TBIOS** is basically a hardware testing utility. If you just built your divIDE you will probably want to flash this. TBIOS source code is available for starting developers as a demonstration of how to interface with ATA devices, how to handle shadow ROM entry points etc. Some goodies are included, such as divIDEo player and dithvIDE picture viewer.

Emulators:

- **SpecEmu** (Windows) - Excellent ZX Spectrum emulator by Mark Woodmass. It has nearly perfect emulation of ULA oddities and complete divIDE emulation is a nice bonus. NOTE: Some antivirus programs may have a false alarm on this file.
- **Spin** (Windows) - one of the best Speccy emulators around that contains complete divIDE emulation.
- **EightyOne** (Windows) - polished Sinclair/Timex emulator by Michael D. Wynne.
- **Fuse** (Linux, Windows) - mature ZX Spectrum emulator which adopted divIDE courtesy of Matthew "Gasman" Westcott.

- **Spediv** (Linux) - enhanced version of [Spectemu](#) by ub880d. Spediv's emulation of memory mapping and ATA layer is well-tested and is able to run any existing divIDE software.
- **Fuse for macOS** (macOS) - macOS port of The Free Unix Spectrum Emulator (Fuse).

Other projects:

- **ZX Spectrum +3e** maintained by Garry Lancaster is an enhancement of the original Amstrad ZX Spectrum models +2A and +3. The +3e ROM supports various ATA interfaces including divIDE.

download

[FATware 0.12](#)

[FATware 0.12a](#) (modified by Velesoft & Micky)

[FATware 0.14 sources](#) for [Zilog Developer Studio 3.68](#)

[FATware 0.14a](#) + boot.bin [14a2](#), [14a3](#) (Modified by Dave)

[TBIOS v1.4 sources](#) for [SjASMPlus](#)

[DithvIDE & Divideo](#) examples with instructions.

[DEMFIIR v81123](#) - unofficial bugfixed version.

technical documentation

[divIDE programming model](#) - description of divIDE interfacing and memory mapping.

[divIDE schematics](#)

[divIDE board layout](#)

[divIDE 5.7c](#) - complete project file of the latest official hardware revision.

[cfIDE 4.7c](#) - complete project file of the CF to IDE adapter (also by Pavel Cimbal).

www.t13.org - valuable resource of ATA/ATAPI related documents including ATA standard drafts.

www.ata-atapi.com - useful information for ATA/ATAPI developers, including sample device driver source code.

www.compactflash.org - official site of the CompactFlash Association.

goodies

DIY how-to - detailed, step-by-step instructions on how to assemble the interface for people who obtained do-it-yourself divIDE kit. Written in friendly language and sent to us courtesy of Alan "Factor6" Petrik.

[logo](#) - download "official" divIDE logo

[small logo](#) - small logo designed for Speccy screen

[SCR logo](#) - small logo in SCR format

[Prototype](#) - image of the first working prototype of divIDE (february 2002)

where to get one

The divIDE is a free project and thus has no official producer. The initial batches of boards produced by Pavel Cimbal were sold mostly as DIY kits at a price of mere 20 EUR (14 GBP), which is basically a cost of components. These are now definitely out of stock and it is unlikely that Pavel himself will produce another batch, mainly due to the lack of time needed for assembling and shipping. Many interfaces (~35 EUR), DIY kits (27 EUR) and various adaptors (4-12 EUR) are produced by Noby, try [his shop](#). There are also some other options.

Check [Sell My Retro](#) shop. There are various sellers who offer divIDE interfaces and accessories

such as CompactFlash cards and various adaptors.

In UK, [RWAP Software](#) offers extended version named divIDE Plus, with custom board layout and changes in firmware have been made to allow the interface to work in 128k BASIC mode.

The [ebay](#) is also good option. From time to time you can find divIDE interfaces there.

disclaimers

The divIDE is merely a hobbyist's project and not a professional design. There may well be bugs, glitches or other unforeseen issues. Most things have been tested quite extensively but no guarantees as to fitness for any particular purpose are offered. Also, it is connected to the user's equipment entirely at their own risk.

The divIDE hardware design, cfiDE hardware design, TBIOS software, fabrication gerbers and schematics supplied remain the copyright of Pavel Cimbal (aka Zilog or Zilogator).

This web design and FATware software remain the copyright of Milos Bazelides (aka Baze).

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Contact Pavel "Zilogator" Cimbal (the hardware author) with any technical questions.

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Contact us with issues related to this page.

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- (1) There are simple interfaces that use 8-bit addressing which virtually reduces the hard drive to half capacity.
 - (2) Compare to ZXATASP (by Samii Vehmaa) using 8255 chip for interfacing. The actual transfer speed here is about 68 KB/sec.
 - (3) Many existing interfaces have compatibility issues as they rely on unspecified ATA behaviour.